parameters.bsn

Purpose

`parameters.bsn` contains the coefficients that apply to the whole watershed: initial soil water, surface runoff lag, peak rate factors, nutrient and pesticide transport coefficients, denitrification, channel routing coefficients, lapse rates, erosion coefficients for sub-daily runs and the starting CO2 concentration. It is named in the second position of the `basin` line of `file.cio` (default `parameters.bsn`). Several of the parameters only matter for the methods selected in `codes.bsn`.

If the file named in `file.cio` does not exist, every parameter keeps its built-in value, after the zero-value replacements described in Zero and missing values.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Line 3 contains 44 values in the order below: 42 real numbers followed by two integers. The reader matches values by position only, so all 44 must be present and in this order. The table gives the column headers the SWAT+ Editor writes; several no longer describe what the model does with the value.

#	Column	Units	Built-in value	Typical range	Description
1	`lai_noevap`	-	3.0	-	Not used. Read and can be changed by calibration, but no routine uses it.
2	`sw_init`	fraction	0.0	0 to 1	Initial soil water as a fraction of field capacity, the same for all soils. 0 means it is calculated from average annual precipitation. See Initial soil water.
3	`surq_lag`	-	4.0	1 to 24	Surface runoff lag coefficient. See Surface runoff lag.
4	`adj_pkrt`	-	1.0	0.5 to 2	Multiplier on the monthly mean fraction of daily rain that falls in the maximum half hour, calculated from the weather generator statistics (limited to 0.1 to 0.95). The daily fraction drawn from it is used by the half-hour intensity peak rate method (`sed_det = 1`), the rainfall erosivity of the MUSLE, urban washoff and the generation of sub-daily rainfall.
5	`adj_pkrt_sed`	-	484.0	-	Peak rate factor of the NRCS peak runoff equation, used for HRU erosion when `sed_det = 0` in `codes.bsn` and for wetlands. See Peak rate factor.
6	`lin_sed`	-	0.0	0 and up	Pesticide groundwater mixing factor. See Pesticides in aquifers.
7	`exp_sed`	-	0.0	0 or 1	Temperature-dependent pesticide decay: `1` on, any other value off. See Pesticide decay and temperature.
8	`orgn_min`	1/day	0.003	0.0003 to 0.003	Rate factor for mineralization of the active humus N and P pools (static carbon model, `carbon = 0`).
9	`n_uptake`	-	20.0	0 to 100	Nitrogen uptake distribution parameter. See Uptake distribution.
10	`p_uptake`	-	20.0	0 to 100	Phosphorus uptake distribution parameter.
11	`n_perc`	-	0.10	0 to 1	Nitrate percolation coefficient: ratio of the nitrate concentration in surface runoff to the concentration in water leaving the top layer.
12	`p_perc`	m^3/Mg	10.0	10 to 17.5	Phosphorus percolation coefficient, used in the leaching of soluble P from each layer.
13	`p_soil`	m^3/Mg	175.0	100 to 200	Phosphorus soil partitioning coefficient: ratio of soluble P in the top 10 mm of soil to soluble P in surface runoff.

#	Column	Units	Built-in value	Typical range	Description
14	p_avail	fraction	0.40	0.01 to 0.7	Phosphorus availability index: fraction of fertilizer P that is in solution after the fast reaction period. Sets the balance between the labile and active mineral P pools in the original P model (soil_p = 0).
15	rsd_decomp	fraction/day	0.05	0.02 to 0.1	Residue decomposition coefficient: daily fraction of residue that decomposes under ideal conditions. Used for every plant whose plnt_decomp in plants.plt is 0.
16	pest_perc	-	0.5	0 to 1	Pesticide percolation coefficient: ratio of the pesticide concentration in surface runoff and lateral flow to the concentration in percolate.
17	msk_co1	-	0.75	0 to 10	Weight of the storage time constant at bankfull depth in the Muskingum storage constant.
18	msk_co2	-	0.25	0 to 10	Weight of the storage time constant at 0.1 bankfull depth in the Muskingum storage constant.
19	msk_x	-	0.20	0 to 0.3	Muskingum weighting factor for inflow and outflow in channel storage.
20	nperco_lchtile	-	0.5	0 to 1	Ratio of the nitrate concentration in tile flow to the concentration in the mobile water of the tile layer.
21	evap_adj	-	0.60	0.5 to 1	Channel evaporation adjustment factor: multiplier on station PET for evaporation from the channel water surface.
22	scoef	-	1.0	0 to 1	Channel storage coefficient of the variable storage routing method (rte_cha = 0 in codes.bsn).
23	denit_exp	-	1.40	0 to 3	Denitrification exponential rate coefficient. See Denitrification.
24	denit_frac	-	1.30	see text	Denitrification threshold on the soil water factor. See Denitrification.
25	man_bact	fraction	0.15	-	Not used.
26	adj_uhyd	h	0.0	0 and up	Added to the base time of the sub-daily unit hydrograph. See unit hydrograph.
27	cn_froz	1/mm	0.000862	-	Frozen soil coefficient of the curve number retention parameter. See Frozen soil.

#	Column	Units	Built-in value	Typical range	Description
28	dorm_hr	h	-1.0	0 to 24	Day length threshold for dormancy. 0 or less means it is calculated from latitude. See Dormancy threshold.
29	plaps	mm/km	0.0	-	Precipitation lapse rate. Positive values increase precipitation with elevation. Used when lapse = 1 in codes.bsn.
30	tlaps	°C/km	6.5	-	Temperature lapse rate. Positive values decrease temperature with elevation. Used when lapse = 1 in codes.bsn.
31	n_fix_max	kg N/ha/day	20.0	1 to 20	Maximum daily nitrogen fixation of a plant.
32	rsd_decay	fraction/day	0.01	0 to 0.05	Minimum daily residue decay fraction. The daily decay rate of residue is never below this value.
33	rsd_cover	ha/t	0.75	-	Residue cover factor in the USLE C factor. See USLE C factor.
34	urb_init_abst	mm	1.0	0 to 10	Maximum initial abstraction on impervious urban areas. Used only with Green-Ampt (gampt = 1).
35	petco_pmpt	%	100.0	-	Not used. The PET of every method is adjusted with pet_co in hydrology.hyd instead.
36	uhyd_alpha	-	1.0	0.5 to 10	Shape coefficient of the gamma unit hydrograph (uhyd = 1 in codes.bsn).
37	splash	-	0.0	0.9 to 3.1	Splash erosion coefficient. See Sub-daily erosion.
38	rill	-	0.0	0.5 to 2	Rill erosion coefficient, a multiplier on the USLE K factor of the top layer.
39	surq_exp	-	0.0	1 to 3	Exponent on bed shear stress in overland flow erosion.
40	cov_mgt	-	0.0	0.001 to 0.45	Scaling factor on the cover and management factor in overland flow erosion.
41	cha_d50	mm	0.0	-	Not used. The median particle size of each channel is read from the channel input.
42	co2	ppm	400.0	-	Atmospheric CO2 concentration, used for every year when there is no co2_yr.dat. See CO2.
43	day_lag_max	days	0	-	Not used as read. The model always sets it to 2.
44	igen	-	5	0 and up	Random number seed code. See Random numbers.

Values are read in one free-format read. A line with fewer values continues onto the next line, and the parameters that are not reached keep their built-in values. A value that cannot be read ends the read without a message.

Zero and missing values

After reading, SWAT+ replaces values that are zero or negative (strictly, below 10^{-6}) for some parameters. For most of them the replacement is the built-in value, so leaving a column at 0 has the same effect as entering the built-in value. These are: `lai_noevap`, `surq_lag`, `adj_pkrt`, `n_uptake`, `p_uptake`, `n_perc`, `p_perc`, `p_soil`, `p_avail`, `rsd_decomp`, `pest_perc`, `msk_co1`, `msk_co2`, `msk_x`, `nperco_lchtile`, `evap_adj`, `denit_exp`, `denit_frac`, `man_bact`, `cn_froz`, `n_fix_max`, `rsd_decay`, `rsd_cover` and `uhyd_alpha`.

The exceptions are:

Column	Value read	Value used
`adj_pkrt_sed`	0 or less	1.0 (the built-in value is 484.0)
`orgn_min`	0 or less	0.0003 (the built-in value is 0.003)
`urb_init_abst`	0 or less	0.0 (the built-in value is 1.0)
`co2`	below 100	400.0
`day_lag_max`	any	2

`sw_init`, `adj_uhyd`, `dorm_hr`, `splash`, `rill`, `surq_exp`, `cov_mgt` and `cha_d50` stay at 0 when 0 is read, and 0 has a meaning for some of them (see the sections below). `lin_sed`, `exp_sed`, `scoef`, `plaps`, `tlaps` and `igen` are used as read.

Initial soil water

All soils start at the same fraction of field capacity. If `sw_init` is 0 or less, the fraction is calculated from the average annual precipitation P (mm) of the HRU's weather generator station:

$$f_{fc} = \frac{P}{P + \exp(9.043 - 0.002135\,P)}$$

A warm-up period of at least one year (`nyskip` in `print.prt`) makes results largely independent of `sw_init`.

Surface runoff lag

In HRUs with a long time of concentration, only part of the surface runoff reaches the channel on the day it is generated. The fraction released each day is

$$f_{rel} = 1 - \exp\left(-\frac{\text{surq_lag}}{t_{conc}}\right)$$

with t_{conc} in hours. In sub-daily runs t_{conc} is expressed in time steps. The rest is kept in surface storage and added to the next day's runoff. Smaller values of `surq_lag` keep more water in storage and smooth the hydrograph.

Peak rate factor

The column labelled `adj_pkrt_sed` by the SWAT+ Editor is the peak rate factor of the NRCS peak runoff equation (see peak runoff rate). The built-in value, 484, is the standard factor for the NRCS dimensionless unit hydrograph. The Editor writes 1.0 in this column, and a value of 0 is also replaced by 1.0. A factor of 1 gives peak rates 484 times lower than the standard factor, which lowers MUSLE sediment yields from HRUs when `sed_det = 0`. The value only affects the peak rate used for erosion and in wetlands; it does not change runoff volumes.

Pesticides in aquifers

The columns labelled `lin_sed` and `exp_sed` by the Editor are pesticide parameters in 62.0.1. In each aquifer, the pesticide concentration in return flow and seepage is the pesticide mass divided by a mixing volume:

$$C = \frac{M_{pest}}{f_{mix}\, S_{aqu} + Q_{rtn} + Q_{revap} + Q_{seep}}$$

where f_{mix} is `lin_sed`, S_{aqu} is the aquifer storage and the Q terms are the day's return flow, revap and seepage (mm). With `lin_sed = 0` the concentration is based on the day's outflows only. Larger values dilute the pesticide in more of the stored water and slow its release.

Pesticide decay and temperature

With `exp_sed = 1` (the value 1.0; other values such as 2 do not count), pesticide decay in soil, on foliage, in aquifers, in channel water and in channel bed sediment uses half-lives adjusted for temperature with a Q10 model:

$$t_{1/2}(T) = t_{1/2,25}\, Q_{10}^{(25-T)/10}, \qquad Q_{10} = 2.58$$

where $t_{1/2,25}$ is the half-life from the pesticide database, taken as the value at 25 °C, and the daily fraction remaining is $\exp(-0.693/t_{1/2}(T))$. Soil decay uses the layer temperature, foliar decay the top soil layer temperature, and channel decay the water temperature. In aquifers the temperature is fixed at 0 °C, which makes aquifer half-lives about 10.7 times longer than the database value. Any other value of `exp_sed` uses the database half-lives without a temperature adjustment.

The Ames example file (see Example) has `exp_sed = 1.0`, so temperature-dependent decay is on in that dataset.

Uptake distribution

The maximum nitrogen uptake from the soil above depth z is

$$U(z) = U_{pot}\, \frac{1 - \exp(-\beta_n\, z/z_{root})}{1 - \exp(-\beta_n)}$$

where β_n is `n_uptake` and z_{root} is the root depth. Larger values concentrate uptake near the surface. Because the top 10 mm interacts with surface runoff, `n_uptake` affects the nitrate available for runoff; lower layers can make up a shortfall in the upper layers, so plant N stress changes little with this parameter. `p_uptake` works the same way for phosphorus.

Denitrification

Denitrification in a soil layer is

$$N_{denit} = NO_3 \left[1 - \exp\left(-\text{denit_exp}\ \gamma_{tmp}\ \frac{OC}{100}\right)\right]$$

where γ_{tmp} is the temperature factor and OC the organic carbon content (%). It only occurs on days when the soil water factor γ_{sw} is at or above `denit_frac`.

With the static carbon model (`carbon = 0`), $\gamma_{sw} = 0.1 + 0.9\sqrt{SW/FC}$, with no upper limit. With the default `denit_frac` of 1.3, denitrification needs $SW/FC \geq 1.78$, that is, a layer well above field capacity.

With the dynamic carbon model (`carbon = 2`), γ_{sw} comes from the water factor selected in `carbon.bsn` and is never above 1. Any `denit_frac` above 1, including the default 1.3, then switches denitrification off for the whole run. Datasets using `carbon = 2` need a `denit_frac` of 1 or less to simulate denitrification.

Frozen soil

When the second soil layer is at or below 0 °C, the curve number retention parameter S is reduced:

$$S_{frz} = S_{max}\,[1 - \exp(-\text{cn_froz} \cdot S)]$$

which increases runoff from frozen soil. The same adjustment is applied with Green-Ampt.

Dormancy threshold

A plant goes dormant when the day length falls below the station's minimum day length plus a threshold t_{dorm} (hours). If `dorm_hr` is above 0, t_{dorm} = `dorm_hr` everywhere. Otherwise it depends on the latitude ϕ of the weather generator station: 1 h above 40°, $(|\phi| - 20)/20$ h between 20° and 40°, and -1 h below 20°.

USLE C factor

The C factor is the product of a residue factor and a biomass factor:

$$C = \exp\left(-\text{rsd_cover} \cdot \frac{R+1}{1000}\right)\,[1 - f_{cov}\,\exp(-0.328\,h_c)]$$

where R is the residue on the surface (kg/ha), h_c is the canopy height (m), and f_{cov} is the ground cover fraction of the above-ground biomass. This is the only C factor method in 62.0.1 (see USLE C factor).

Sub-daily erosion

With Green-Ampt (`gampt` above 0 in `codes.bsn`), sediment from each time step is the sum of splash and overland flow erosion. Splash erosion decreases with runoff depth h (mm) as $\exp(-\text{splash} \cdot h/1000)$ and depends on rainfall kinetic energy and a soil texture erodibility. Overland flow erosion is

$$E_{ov} = 11.02\ \text{rill}\ K\ \text{cov_mgt}\ C\ \tau^{\,\text{surq_exp}}$$

in kg/h/m², where K is the USLE K factor of the top layer, C is a cover factor from biomass and residue, and τ is the bed shear stress (N/m²). With the built-in values of 0 for `rill` or `cov_mgt`, overland flow erosion is zero.

CO2

Without a `co2_yr.dat` file in the run folder, every year uses `co2`. With the file, the yearly values come from it (see co2_yr.dat). The radiation use efficiency of plants is adjusted for CO2 only when `co2` in `parameters.bsn` is above 350 ppm, even when the yearly values come from `co2_yr.dat`. Penman-Monteith PET uses the yearly values in all cases.

Random numbers

With `igen = 0`, the weather generator and other random processes start from fixed seeds, so every run gives the same random sequence. With any other value, each seed is advanced by a number of steps that depends on `igen`, so different `igen` values give different sequences. The same `igen` always gives the same sequence, so a new value is needed for each different realisation.

Calibration

These parameters can be changed during calibration through `calibration.cal` with object type `bsn`, using their model names: `surlag` (`surq_lag`), `adj_pkr` (`adj_pkrt`), `prf` (`adj_pkrt_sed`), `evlai` (`lai_noevap`), `ffcb` (`sw_init`), `cmn` (`orgn_min`), `nperco` (`n_perc`), `pperco` (`p_perc`), `phoskd` (`p_soil`), `psp` (`p_avail`), `rsdco` (`rsd_decomp`), `percop` (`pest_perc`), `msk_co1`, `msk_co2`, `msk_x`, `nperco_lchtile`, `evrch` (`evap_adj`), `cdn` (`denit_exp`), `sdnco` (`denit_frac`), `tb_adj` (`adj_uhyd`), `n_updis` (`n_uptake`), `rsd_covco` (`rsd_cover`), `dorm_hr`, `plaps` and `tlaps`. See parameter change files.

The changes in calibration.cal are applied once, after the HRUs, channels and weather stations have been set up. Several of these parameters are only used during that setup, so changing them in calibration.cal has no effect on the run:

- cmn, nperco, pperco, phoskd, psp and nperco_lchtile have already been copied to each HRU. Use the HRU-level names cmn_h, nperco_h, pperco_h, phoskd_h, psp_h and nperco_lchtile_h instead.
- ffcb: the initial soil water has already been set.
- adj_pkr and dorm_hr: the half-hour rainfall fractions and the dormancy thresholds of the weather stations have already been calculated.
- msk_co1, msk_co2 and msk_x: the Muskingum coefficients of the channels have already been calculated.
- rsdco: it has already been copied to the plants with plnt_decomp = 0.

surlag is recalculated into the runoff release fraction when it is changed, and the others are used directly during the simulation, so their changes take effect.

Example

refdata/Ames_sub1/parameters.bsn in the SWAT+ repository:

```
parameters.bsn: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
  lai_noevap       sw_init      surq_lag      adj_pkrt  adj_pkrt_sed       lin_sed       exp_sed      orgn
     3.00000       0.50000       4.00000       1.00000       1.00000       0.00000       1.00000       0.0
```

How the model reads some of these values:

- sw_init = 0.5: soils start at half of field capacity.
- adj_pkrt_sed = 1.0: the NRCS peak rate factor is 1. The standard factor is 484.
- exp_sed = 1.0: temperature-dependent pesticide decay is on.
- orgn_min = 0.0: replaced by 0.0003.
- cn_froz = 0.0: replaced by 0.000862.
- dorm_hr = 0.0: the dormancy threshold is calculated from latitude.
- plaps = 0.0 and tlaps = 0.0: no lapse rate adjustment, even if lapse = 1 in codes.bsn.
- denit_frac = 1.3: this dataset uses carbon = 2 in its codes.bsn, so no denitrification occurs (see Denitrification).
- petco_pmpt, cha_d50 and man_bact have no effect; day_lag_max is set to 2.
- igen = 0: fixed random seeds.

Related files

- file.cio: names parameters.bsn on its basin line.
- codes.bsn: method codes that decide which of these parameters are used.
- hydrology.hyd: HRU-level hydrology parameters, including pet_co.
- plants.plt: plnt_decomp overrides rsd_decomp for a plant.
- Parameter change files: calibration names for these parameters.
- Nutrient cycling, pesticides, erosion and sediment, channels and hydrology: process descriptions.

Sources

- Reader: basin_read_prm.f90, followed by the zero-value replacements in basin_prm_default.f90; both called from proc_bsn.f90
- Type definition: type basin_parms in basin_module.f90
- File name slot: in_basin%parms_bas in input_file_module.f90
- Copy to HRUs: hrudb_init.f90
- Hydrology: hydro_init.f90, time_conc_init.f90, sq_dailycn.f90, sq_greenampt.f90, unit_hyd.f90, cli_lapse.f90
- Weather generator: cli_initwgn.f90, gcycl.f90, co2_read.f90
- Peak rate and erosion: ero_pkq.f90, wetland_control.f90, ero_cfactor.f90, ero_ovrsed.f90
- Pesticides: aqu_1d_control.f90, pest_decay.f90, pest_tempdecay_module.f90, ch_rtpest.f90, pest_lch.f90

- Nutrients: nut_nminrl.f90, cbn_zhang2.f90, nut_nlch.f90, nut_solp.f90, nut_pminrl.f90, rsd_decomp.f90, cbn_surfrsd_decomp.f90, plantparm_init.f90
- Plants: pl_nup.f90, pl_pup.f90, pl_nfix.f90, pl_biomass_gro.f90
- Channels: sd_hydsed_init.f90, ch_rtmusk.f90, ch_rthr.f90, ch_rtday.f90
- Calibration: cal_parm_select.f90

Column	Fortran variable
lai_noevap	bsn_prm%evlai
sw_init	bsn_prm%ffcb
surq_lag	bsn_prm%surlag
adj_pkrt	bsn_prm%adj_pkr
adj_pkrt_sed	bsn_prm%prf
lin_sed	bsn_prm%pestgwfact
exp_sed	bsn_prm%temp_decay
orgn_min	bsn_prm%cmn
n_uptake	bsn_prm%n_updis
p_uptake	bsn_prm%p_updis
n_perc	bsn_prm%nperco
p_perc	bsn_prm%pperco
p_soil	bsn_prm%phoskd
p_avail	bsn_prm%psp
rsd_decomp	bsn_prm%rsdco
pest_perc	bsn_prm%percop
msk_co1, msk_co2, msk_x	bsn_prm%msk_co1, %msk_co2, %msk_x
nperco_lchtile	bsn_prm%nperco_lchtile
evap_adj	bsn_prm%evrch
scoef	bsn_prm%scoef
denit_exp	bsn_prm%cdn
denit_frac	bsn_prm%sdnco
man_bact	bsn_prm%bact_swf
adj_uhyd	bsn_prm%tb_adj
cn_froz	bsn_prm%cn_froz
dorm_hr	bsn_prm%dorm_hr
plaps, tlaps	bsn_prm%plaps, %tlaps
n_fix_max	bsn_prm%nfixmx
rsd_decay	bsn_prm%decr_min
rsd_cover	bsn_prm%rsd_covco
urb_init_abst	bsn_prm%urb_init_abst
petco_pmpt	bsn_prm%petco_pmpt
uhyd_alpha	bsn_prm%uhalpha
splash	bsn_prm%eros_spl
rill	bsn_prm%rill_mult
surq_exp	bsn_prm%eros_expo
cov_mgt	bsn_prm%c_factor
cha_d50	bsn_prm%ch_d50
co2	bsn_prm%co2
day_lag_max	bsn_prm%day_lag_mx
igen	bsn_prm%igen